

Activation functions

Project: Image-pattern Network Investigation

1. Your mission

Your learning target

I can trace identity ReLU and sigmoid on a numerical score.

I can explain how a nonlinear activation changes a computation.

I can distinguish an activation output from a final class or verified probability.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Bias terms and decision boundaries. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Representing inputs as vectors. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Trace identity ReLU and sigmoid on a numerical score

Explain how a nonlinear activation changes a computation

Distinguish an activation output from a final class or verified probability

Retrieve what you already know

For $z = 2 \text{ times width} - \text{height} - 3$ at $[3, 3]$, what are the score and class under $z \geq 0$?

My starting explanation

This week's end goal

By the end, I can explain activation functions using a traced example and a stated limitation.

2. Notice the evidence

First compute an affine score $z = w \cdot x + b$. An activation then returns $a = f(z)$. In a network this value may become an input to another layer; it is not automatically a class prediction. We will compare three functions using the same incoming scores so that only the activation changes.

Identity returns z unchanged. ReLU, pronounced ree-loo, returns the larger of zero and z . Thus $\text{ReLU}(-2) = 0$, $\text{ReLU}(0) = 0$ and $\text{ReLU}(2) = 2$. Positive scores keep their magnitude; ReLU does not turn every positive value into 1.

Incoming z · Identity output · ReLU output

Incoming z: -2

Identity output: -2

ReLU output: 0

Incoming z: 0

Identity output: 0

ReLU output: 0

Incoming z: 2

Identity output: 2

ReLU output: 2

These are fixed arithmetic functions. Selecting a function or changing its input is not itself a training procedure. Later weeks will introduce losses and parameter updates.

List the given facts and one fact that is not supplied.

3. See how it works

Sigmoid is $1 / (1 + \exp(-z))$. Here $\exp(t)$ means e raised to power t , where e is approximately 2.718. A calculator or the supplied Python code can evaluate it; no calculus is required this week. For $z = -2, 0$ and 2 , sigmoid outputs are approximately 0.1192, 0.5 and 0.8808.

For every finite mathematical input, sigmoid is strictly between 0 and 1, though computer rounding can produce an endpoint for extreme scores. Negative inputs produce values below 0.5, zero gives 0.5, and positive inputs give values above 0.5. A class rule still needs to be stated: if class 1 means $\text{sigmoid}(z) \geq 0.5$, these three predictions are 0, 1 and 1.

```
from math import exp

def sigmoid(z):
    if z >= 0:
        return 1 / (1 + exp(-z))

    e = exp(z)
    return e / (1 + e)

for z in [-2, 0, 2]:
    print(z, max(0, z), round(sigmoid(z), 4))
```

The two algebraically equivalent branches avoid taking $\exp$ of a large positive number. A value within $[0, 1]$ is not automatically a reliable probability: its interpretation depends on the fitted task and evaluation evidence.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

Use the lesson sigmoid values for $-2, 0, 2$. Predict classes under a ≥ 0.5 , then explain the zero-score case.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: Replace ReLU with identity.

Prediction: h becomes -2, 0, 1 and outputs become -4, 2, 5.

Reason: Identity retains the negative intermediate value instead of clipping it.

5. Words that unlock the idea

Word · Meaning

Word: Activation function

Meaning: A function applied to a computed score

Word: ReLU

Meaning: The function $\max(0, z)$, retaining positive scores and replacing negative scores with zero

Word: Sigmoid

Meaning: The function $1 / (1 + \exp(-z))$, mapping finite real scores into the interval between 0 and 1

Word: Nonlinear function

Meaning: A function that cannot be represented by one fixed slope and offset over its entire domain

Use two terms to explain a step in this week's example.

6. Follow a worked example

Three functions receive the same score

Same incoming score · Apply one function · Activation output

Same incoming score: z is -2

Apply one function: Identity of z

Activation output: -2

Same incoming score: z is -2

Apply one function: ReLU of z

Activation output: 0

Same incoming score: z is -2

Apply one function: Sigmoid of z

Activation output: About 0.1192

Each row starts again from the same score -2 and applies one function. The functions are alternatives here, not successive steps. These outputs are activation values, not class labels.

For $z = -1$ and $z = 1$, give identity and ReLU outputs. Using $\text{sigmoid}(-1) = 0.2689$ and $\text{sigmoid}(1) = 0.7311$, give classes under the rule $a \geq 0.5$.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Activation functions

See what ReLU removes before the next weighted step.

The facts we will use

Toy network $z=x-3$; $h=\max(0, z)$; $y=3h+2$. Inputs $x=1, 3, 4$.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/4

ReLU hidden computation at $x=1$

Input 1 x weight 1 = 1

Add the products and bias -3: total = -2

ReLU: use $\max(0, \text{total}) \rightarrow \text{output } 0$

Activation functions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute preactivation

z values are -2, 0, 1.



2. Apply ReLU

h values are 0, 0, 1; the negative value is replaced by 0.



3. Compute output

y values are 2, 2, 5; x = 1 and x = 3 collapse to the same output.

CHANGE ONE THING

Replace ReLU with identity.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Compute preactivation

What happens and why: z values are -2, 0, 1.

Step: 2. Apply ReLU

What happens and why: h values are 0, 0, 1; the negative value is replaced by 0.

Step: 3. Compute output

What happens and why: y values are 2, 2, 5; $x = 1$ and $x = 3$ collapse to the same output.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

ReLU means keep a positive value and replace a negative one with zero. The output layer receives that transformed value, not the original input. Inputs one and three therefore both deliver zero to the final layer and produce two. Replacing ReLU with identity changes the procedure: negative intermediate values now remain available to the next layer.

What this example does not establish

ReLU is one activation choice. Clipping negatives does not itself train the network or prove useful representation.

Read the labelled picture: Which original inputs cannot be distinguished by this network's final output? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Change one thing: Replace ReLU with identity. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

An affine function has one fixed slope. ReLU has slope zero on the negative side and slope one on the positive side, so no single straight line fits the whole function. For a concrete check, inputs -2, 0, 2 give outputs 0, 0, 2. The average of the endpoint outputs is 1, but the output at the midpoint 0 is 0. This differs from an affine function, which preserves midpoints.

For a one-input chain with identity between stages, $h = 2x + 1$ followed by $y = 3h - 2$ collapses to $y = 6x + 1$. Insert ReLU instead: $h = \max(0, 2x + 1)$, $y = 3h - 2$. At $x = -1$, the identity chain gives -5 while the ReLU chain gives -2. At $x = 1$, both give 7. The nonlinear step can change the overall family of functions.

Locate the earliest unsupported step

Compare ReLU at inputs -2, 0, 2. Give the average of the two endpoint outputs and the output at the midpoint; explain what this shows.

State one repair and the checks you would rerun.

8. Practise together

For scores -2, 0, 2, list identity and ReLU outputs. Explain why $\text{ReLU}(2)$ is not 1.

For $z = -1$ and $z = 1$, give identity and ReLU outputs. Using $\text{sigmoid}(-1) = 0.2689$ and $\text{sigmoid}(1) = 0.7311$, give classes under the rule $a \geq 0.5$.

Explain your prediction to a partner before comparing results.

Practise with less help: Apply a declared class rule

Use workbook W2 for the connected example “Activation functions”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Activation functions. Proposed explanation: "ReLU returns a class label". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which original inputs cannot be distinguished by this network's final output? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

Use the lesson sigmoid values for -2, 0, 2. Predict classes under a ≥ 0.5 , then explain the zero-score case.

```
from math import exp

def sigmoid(z):
    if z >= 0:
        return 1 / (1 + exp(-z))
    e = exp(z)
    return e / (1 + e)

for z in [-2, 0, 2]:
    print(z, max(0, z), round(sigmoid(z), 4))
```

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

For scores -2, 0, 2, list identity and ReLU outputs. Explain why $\text{ReLU}(2)$ is not 1.

Compare ReLU at inputs -2, 0, 2. Give the average of the two endpoint outputs and the output at the midpoint; explain what this shows.

A model returns a sigmoid value 0.9 on one case. Does the numerical range alone establish a calibrated 90 percent probability? Explain what needs checking.

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

Add a labelled activation functions evidence sheet to Image-pattern Network Investigation. Show how you can trace identity relu and sigmoid on a numerical score. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

A nonlinear activation allows a network to represent relationships that a chain of affine operations alone

cannot express. It does not guarantee that training will find a useful model or that more layers will help. ReLU can erase differences between negative scores; sigmoid can compress very large scores into nearly identical values. The architecture, data and training procedure still matter.

A learner reports $\text{sigmoid}(2)$ is about 0.8808 and concludes the system will be correct 88 percent of the time for every future case. What does the calculation establish, and what evidence is still missing?

Next week, we will compose layers while keeping every intermediate value and its role visible.

A model returns a sigmoid value 0.9 on one case. Does the numerical range alone establish a calibrated 90 percent probability? Explain what needs checking.

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

Three functions receive the same score

Same incoming score · Apply one function · Activation output

Same incoming score: z is -2

Apply one function: Identity of z

Activation output: -2

Same incoming score: z is -2

Apply one function: ReLU of z

Activation output: 0

Same incoming score: z is -2

Apply one function: Sigmoid of z

Activation output: About 0.1192

Each row starts again from the same score -2 and applies one function. The functions are alternatives here, not successive steps. These outputs are activation values, not class labels.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Trace a new chain independently

Let $z = x - 2$, $a = \text{ReLU}(z)$, $y = 2a + 1$. For $x = 0, 2, 5$, record every z , a and y . Then replace ReLU with identity and give all three y values. Which inputs have the same ReLU output, and why?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Let $z = x - 2$, $a = \text{ReLU}(z)$, $y = 2a + 1$. For $x = 0, 2, 5$, record every z , a and y . Then replace ReLU with identity and give all three y values. Which inputs have the same ReLU output, and why?

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: See what ReLU removes before the next weighted step.

By the end, I can · How I will show it

By the end, I can: Trace identity ReLU and sigmoid on a numerical score

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Explain how a nonlinear activation changes a computation

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Distinguish an activation output from a final class or verified probability

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Trace identity ReLU and sigmoid on a numerical score

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.